CONCRETE TECHNOLOGY

Subject Code : 23A01402T | Handwritten Study Notes

UNIT – I

CEMENTS & AGGREGATES

Syllabus covered in this unit

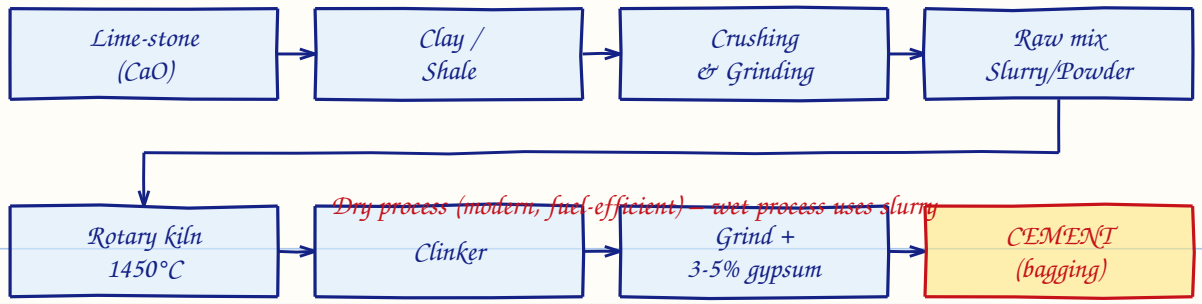
Portland cement – chemical composition – hydration – setting – fineness – structure of hydrated cement – physical tests – grades of cement – admixtures (mineral & chemical) – classification of aggregates – shape, texture, bond, strength – specific gravity, bulk density, porosity, absorption, moisture – bulking – deleterious substances – soundness – alkali-aggregate reaction – thermal properties – sieve analysis – FM – grading – max size – mixing water

1. PORTLAND CEMENT

Cement = a **finely ground hydraulic binder**; when mixed with water it sets and hardens by chemical reaction (hydration) – even under water. Joseph Aspdin (1824) patented it & named it "Portland" cement because it resembled Portland stone.

1.1 Manufacture (brief)

- Raw materials : **calcareous** (limestone, chalk) + **argillaceous** (clay, shale) + gypsum (added at the end).
- Proportion about 3 : 1 (lime : clay) – burnt in a rotary kiln at about 1400–1450 °C → **clinker** (nodules 3–25 mm).
- Clinker is cooled, ground with **3–5 % gypsum** to control the setting of C_3A.



1.2 Chemical composition (oxide composition)

Oxide	Per cent	Function / remark
Lime CaO	60 – 67	Main ingredient; excess free lime → unsoundness; deficiency → low strength, quick set
Silica SiO_2	17 – 25	Gives strength (C_3S, C_2S)

Oxide	Per cent	Function / remark
Alumina Al_2O_3	3 – 8	Quick setting, lowers clinkering temp.
Iron oxide Fe_2O_3	0.5 – 6	Colour, helps fusion
Magnesia MgO	0.1 – 4	Excess (>6 %) → unsoundness
Sulphur trioxide SO_3	1 – 3	Excess → unsoundness (sulphate attack)
Alkalis $Na_2O + K_2O$	0.4 – 1.3	Excess → efflorescence, alkali-aggregate reaction

- IS 269 limits : insoluble residue ≤ 4 %, MgO ≤ 6 %, loss on ignition ≤ 5 %, SO_3 ≤ 2.5 % (C_3A ≤ 5 %) or 3 % (C_3A > 5 %), chloride ≤ 0.1 %.
- **Lime Saturation Factor** $LSF = (CaO - 0.7\ SO_3) / (2.8\ SiO_2 + 1.2\ Al_2O_3 + 0.65\ Fe_2O_3)$ → 0.66 to 1.02. **Alumina–iron ratio** = Al_2O_3/Fe_2O_3 ≥ 0.66.

1.3 Bogue's compounds (main compounds of cement)

Compound	Formula (short)	%	Behaviour
Tricalcium silicate C_3S	$3CaO{\cdot}SiO_2$	25–50	Hydrates fast, high heat (500 J/g); early strength (7 days) & final set
Dicalcium silicate C_2S	$2CaO{\cdot}SiO_2$	20–45	Slow; low heat (260 J/g); later strength after 28 days
Tricalcium aluminate C_3A	$3CaO{\cdot}Al_2O_3$	5–11	Flash set; highest heat (910 J/g); poor sulphate resistance
Tetra-calcium alumino-ferrite C_4AF	$4CaO{\cdot}Al_2O_3{\cdot}Fe_2O_3$	6–12	Colour (grey); low heat (420 J/g); little strength

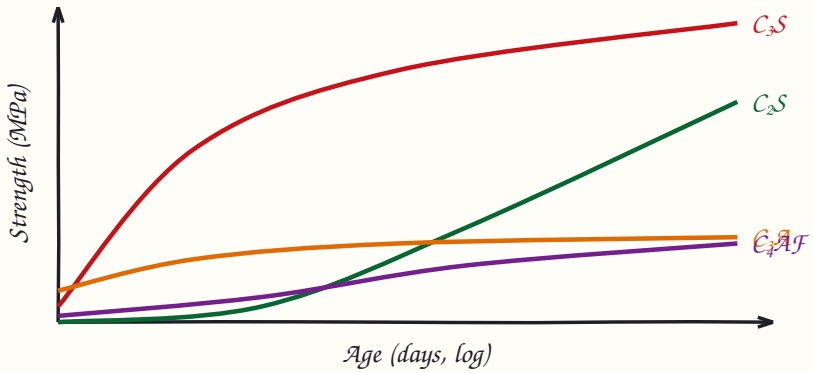
Bogue's equations (when $A/F \geq 0.66$) $C = \%CaO$, $S = \%SiO_2$, $A = \%Al_2O_3$, $F = \%Fe_2O_3$

$C_3S = 4.07\ C - 7.60\ S - 6.72\ A - 1.43\ F - 2.85\ SO_3$

$C_2S = 2.87\ S - 0.754\ C_3S$

$C_3A = 2.65\ A - 1.69\ F$

$C_4AF = 3.04\ F$



- C_3S : early strength
- C_2S : later strength
- C_3A : flash set, heat, little strength
- C_4AF : colour, little strength

2. HYDRATION OF CEMENT

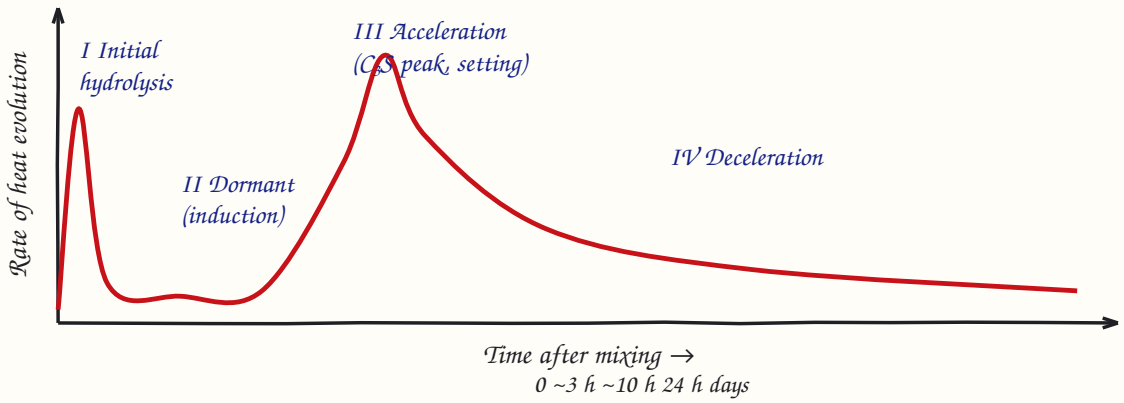
Hydration = chemical reaction between cement compounds and water forming **hydration products (gel)** that bind the aggregates. It is **exothermic**.

- $2C_3S + 6H \rightarrow C_3S_2H_3$ (C-S-H gel) + $3Ca(OH)_2$ [100 : 24 → 75 : 49 by weight]
- $2C_2S + 4H \rightarrow C_3S_2H_3 + Ca(OH)_2$ (less lime, hence more durable)
- $C_3A + 6H \rightarrow C_3AH_6$ (very fast, flash set if no gypsum)
- $C_3A + 3CaSO_4 \cdot 2H_2O + 26H \rightarrow$ **ettringite** ($C_3A \cdot 3CaSO_4 \cdot 32H_2O$) → later converts to monosulphate.
- $C_4AF + CaSO_4 + H \rightarrow$ calcium sulpho-aluminoferrite + CH + hydrates.

Remember ★

- Water needed for complete hydration ≈ **23 %** (chemically bound) + **15 %** (gel water) ≈ **38 %** by weight of cement → minimum w/c ≈ 0.38.
- $Ca(OH)_2$ (CH) gives alkalinity (pH ≈ 12.5) which protects steel but is weak & soluble (leaching).
- Heat of hydration : 7 days ≈ 250–300 J/g, 28 days ≈ 310–350 J/g → problem in mass concrete.

2.1 Stages of hydration (heat evolution curve)



- Stage I – rapid dissolution, C_3A & gypsum react (ettringite film) – lasts few minutes.
- Stage II – **dormant / induction period** (1–3 h) : paste stays plastic → time for transport & placing.
- Stage III – acceleration : C_3S & C_2S hydrate fast, **initial & final set** occur, CH crystallises.
- Stage IV/V – deceleration, diffusion controlled, slow continued strength gain for months.

2.2 Setting of cement

- **Setting** = change from fluid to rigid state (no strength yet); **hardening** = gain of strength after setting.

- **Initial set** : paste starts losing plasticity (≥ 30 min as per IS) ; **final set** : paste fully rigid, can bear some load (≤ 600 min).
- **Flash set** : C_3A reacts instantly with water – stiffening with lot of heat, can't be reworked; prevented by gypsum.
- **False set** : abnormal premature stiffening without much heat (dehydration of gypsum to hemi-hydrate); regained on remixing without extra water.
- Factors : cement composition (C_3A), fineness, w/c, temperature (high → faster), gypsum content, admixtures.

3. FINENESS OF CEMENT

- Fineness controls **rate of hydration** : finer cement → more surface → faster set, early strength, more heat, more shrinkage; but coarse particles (>45 μm) may never fully hydrate.
- Typical OPC : specific surface 300–400 m^2/kg (IS 269 minimum 225 m^2/kg).

Methods of determining fineness

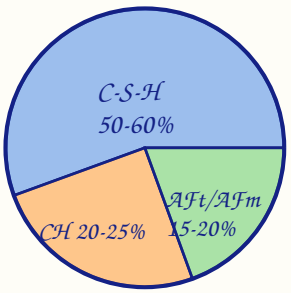
1. **Sieve test** (IS 4031 Part 1) – 100 g cement on 90 μm IS sieve, sieved 15 min; residue should be ≤ 10 %.
2. **Air permeability (Blaine's) method** – specific surface from time for fixed air volume to flow through a cement bed of known porosity : $S \propto \sqrt{(e^3)} \cdot \sqrt{t} / \rho(1-e)$.
3. **Wagner turbidimeter** – sedimentation of particles in kerosene, light transmission gives particle-size distribution.
4. **Lea & Nurse permeability apparatus**, laser diffraction (modern).

Point ★

Finer cement → greater **bleeding control**, better workability, but more **heat & shrinkage**, and faster air-setting (deterioration in storage).

4. STRUCTURE OF HYDRATED CEMENT PASTE

- Hydrated paste = **C-S-H gel** (tobermorite-like, poorly crystalline, 50–60 % of solids, specific surface 200–300 m^2/g) + **$Ca(OH)_2$** crystals (20–25 %) + **sulpho-aluminates** (AFt, AFm 15–20 %) + unhydrated clinker + pores.
- Strength arises from **Van der Waals' forces** (physical bonding) between gel particles; gel has very large surface.



Solids of hydrated paste

Pores in hydrated paste :

- Gel pores ≈ 0.5 – 2.5 nm (inside C-S-H)
- Capillary pores ≈ 10 nm – 50 µm (space left by excess mix water)
- Air voids : entrapped (>1 mm) / entrained (50 µm – 1 mm)

Low w/c → fewer capillary pores → higher strength

Types of water in the paste (Powers–Brunauer model)

- **Chemically combined water** (non-evaporable) – part of hydrates; lost only at high temp.
- **Gel (interlayer) water** – adsorbed on gel surface, lost on drying below ~11 % RH; causes **creep & shrinkage**.
- **Capillary water** – free water in capillary pores (>50 nm): removal causes little shrinkage.
- **Free / evaporable water** – in large voids; evaporates easily.

Powers' relations (gel/space ratio) – used again in Unit III

Gel/space ratio $X = 0.657\,\alpha / (0.319\,\alpha + w/c)$ Strength $f = 240\,X^3$ (MPa)

α = degree of hydration; gel volume = 2.06 × volume of cement hydrated

5. TESTS FOR PHYSICAL PROPERTIES (IS 4031)

Test	Procedure / result
1. Fineness	Sieve test (90 µm, ≤ 10 %) / Blaine (≥ 225 m^2/kg)
2. Standard consistency	**Vicat apparatus**, plunger 10 mm dia, 300 g. Water % (P) at which plunger stops **5–7 mm from bottom** (33–35 mm penetration). Normal P ≈ 26–33 %.
3. Initial setting time	Vicat with **1 mm² square needle**; water = 0.85 P; time when needle stops **5 ± 0.5 mm from base**. ≥ 30 min.
4. Final setting time	Needle with **annular collar (5 mm)** : needle makes an impression but collar doesn't. ≤ 600 min.
5. Soundness	**Le Chatelier** (mould 30 mm × 30 mm, split cylinder, arms 165 mm) : water 0.78 P, 24 h curing, boil 3 h → expansion ≤ **10 mm**. **Autoclave** test ≤ 0.8 %.
6. Compressive strength	Mortar 1 : 3 (cement : standard sand, IS 650), water = (P/4 + 3.0) %, **70.6 mm cubes**; tested at 3, 7, 28 days.
7. Specific gravity	Le Chatelier flask, kerosene ; **3.10 – 3.15**.
8. Heat of hydration	Calorimeter ; for low heat cement 7 d ≤ 272 kJ/kg.

Test	Procedure / result
9. Tensile / Soundness extras	Briquette test (obsolete) ; Setting & expansion due to MgO, free lime, SO_3.

Standard consistency – Vicat ★

Plunger : 10 mm dia × 50 mm long, mass 300 g. Mould : 40 mm deep, 80–90 mm dia. Trials of 300 g cement + 25 %, 28 %, 30 % … water till plunger rests **5–7 mm above the base**.

6. GRADES AND TYPES OF CEMENT

6.1 Grades of OPC (IS 269 / 8112 / 12269)

Grade (28-day strength)	3-day (MPa)	7-day (MPa)	28-day (MPa)
33 grade (IS 269)	≥ 16	≥ 22	≥ 33
43 grade (IS 8112)	≥ 23	≥ 33	≥ 43
53 grade (IS 12269)	≥ 27	≥ 37	≥ 53

- 33 – plastering, masonry ; 43 – general RCC ; 53 – high-rise, prestressed, precast, roads, high-strength concrete.
- Grade number = minimum 28-day compressive strength of 70.6 mm mortar cubes in N/mm^2.

6.2 Other cements

Type (IS code)	Salient features / use
Rapid Hardening (IS 8041)	More C_3S, finer; 3-day strength = OPC 7-day; early de-shuttering, repairs, cold weather
Low Heat (IS 12600)	Less C_3A (<6 %) & C_3S, more C_2S; dams, mass concrete
Sulphate Resisting (IS 12330)	C_3A ≤ 5 %; marine works, sewers, foundations in sulphate soils
Portland Pozzolana PPC (IS 1489)	15–35 % fly ash (Part 1) / calcined clay (Part 2); low heat, better durability, slow early strength
Portland Slag PSC (IS 455)	25–70 % GGBS; low heat, sulphate & chloride resistance, marine works
White / Coloured (IS 8042)	Low iron; decorative, floor finishes, architectural
High Alumina (IS 6452)	35–40 % alumina; very early strength, refractory; not for structural use due to conversion
Hydrophobic (IS 8043)	Water-repellent film on grains; long storage, monsoon
Oil-well, Expansive, Masonry (IS 3466)	special uses

7. ADMIXTURES

Admixture = material other than cement, water, aggregates added **immediately before or during mixing** to modify properties of fresh / hardened concrete (IS 9103, ASTM C494). Used ≤ 5 % by wt. of cement.

7.1 Chemical admixtures

Type	Examples	Effect / dosage
Accelerators	$CaCl_2$ (≤ 1.5–2 %), triethanolamine, sodium silicate, calcium formate	Fast set & early strength; cold weather. **$CaCl_2$ not for RCC** (corrosion), increases shrinkage
Retarders	Sugar, gypsum, calcium sulphate, starch, lignosulphonates, hydroxy-carboxylic acids	Delay setting, reduce heat; hot weather, long haul RMC, large pours
Air-entraining	Vinsol resin, vegetable oils/fats, fatty acids, abietic acid salts, detergents	3–6 % tiny (50 µm) bubbles; better freeze–thaw, workability, less bleeding; **strength drops ≈ 5 % per 1 % air**
Plasticizers (water reducers)	Lignosulphonates, hydroxylated carboxylic acids	5–15 % water reduction; dose 0.1–0.4 %
Super-plasticizers (HRWR)	SNF (naphthalene), SMF (melamine), modified lignosulphonate, **PCE (polycarboxylate)**	20–30 % (PCE up to 40 %) water cut or flowing concrete (slump 150→ 250 mm); dose 0.5–2 %; slump loss in 30–60 min

- Mechanism of plasticizers : adsorb on cement grains → **deflocculation** by electrostatic (SNF/SMF) or **steric** (PCE) repulsion → trapped water released.
- Other : VMA, shrinkage-reducing, corrosion inhibitors, waterproofing, bonding, gas-forming (aluminium powder), pumping aids.

7.2 Mineral admixtures (supplementary cementitious materials)

- **Pozzolana** = siliceous/alumino-siliceous material which by itself has no cementing value but, finely divided and with moisture, reacts with $Ca(OH)_2$ at ordinary temperature → C-S-H : $CH + S + H \rightarrow$ **C-S-H** (secondary gel).

Material	Properties	Effect on concrete
Fly ash (IS 3812)	Residue of pulverised coal, spherical glassy 1–100 µm; Class F (siliceous, $SiO_2+Al_2O_3+Fe_2O_3 \geq 70$ %), Class C (calcareous, self-cementing). Replace 15–35 %	Better workability (ball-bearing effect), low heat, durability, sulphate & ASR resistance; slower early strength, needs longer curing

Material	Properties	Effect on concrete
Silica fume (IS 15388)	By-product of Si / Fe-Si alloy; SiO_2 > 85–90 %; particle size ≈ 0.1 µm (100× finer than cement); 15 000–25 000 m^2/kg; 5–10 %	Filler + highly reactive pozzolan → very dense, 80–100+ MPa HPC; reduces bleeding; high water demand → needs superplasticizer
GGBS	Granulated blast-furnace slag, 25–70 %	Low heat, chloride & sulphate resistance
Rice husk ash, metakaolin	Reactive silica	HPC, durability

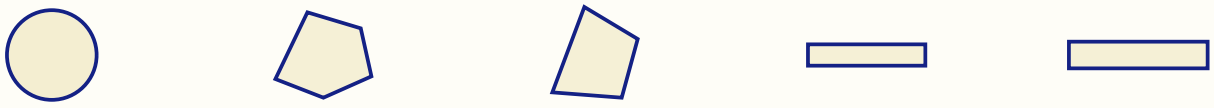
8. AGGREGATES

Aggregates occupy **60–75 % of concrete volume** (70–80 % of mass) – they affect strength, durability, workability, economy, dimensional stability.

8.1 Classification

- **By size** : **Coarse** aggregate – retained on 4.75 mm sieve (nominal 10, 12.5, 20, 25, 40 mm). **Fine** aggregate – passes 4.75 mm, retained on 75 µm (sand). **All-in aggregate** – mix of both.
- **By source / origin** : natural (river, pit, sea, crushed rock – granite, basalt, quartzite, limestone) – artificial (broken brick, slag, clinker, expanded clay).
- **By unit weight** : **Normal weight** (2400 kg/m^3 concrete; bulk density 1520–1680 kg/m^3); **Light-weight** (< 1120 kg/m^3, pumice, vermiculite, expanded shale); **Heavy-weight** (> 2080 kg/m^3; barytes, magnetite – radiation shielding).
- **By shape** : rounded, irregular, angular, flaky, elongated.
- **By geological (petrological) group** : igneous, sedimentary, metamorphic.

9. PARTICLE SHAPE AND TEXTURE



Rounded Irregular Angular Flaky Elongated

- **Rounded** (river gravel) : minimum voids, best workability, poor bond, lower flexural strength.
- **Angular / crushed** : more interlock & bond → higher strength (10–15 % more flexural), but needs ~10 kg more water per m^3; harsh mix.
- **Flaky** (thickness < 0.6 × mean size) & **elongated** (length > 1.8 × mean size) → poor workability, weaker, lie flat with water pockets → limit **combined index ≤ 30–35 %**.
- **Surface texture** : smooth (glassy, chert) ↔ rough (honeycombed, granular) → rough texture gives

better **bond** & strength; smooth gives workability.

Shape tests (IS 2386 Part 1) ★

- **Flakiness index** = (weight passing thickness-gauge slot / total weight) × 100.
- **Elongation index** = (weight retained on length gauge / total) × 100.
- **Angularity number** = 67 − % solid volume of aggregate in a compacted cylinder (0 for rounded, up to 11 for very angular); **Angularity index** = ratio of void % .

10. BOND, STRENGTH & MECHANICAL PROPERTIES

- **Bond** between aggregate and paste (interfacial transition zone, ITZ – weakest link, thickness 15–50 µm) depends on texture, shape, mineralogy, cleanliness. Bond is evaluated indirectly – by flexural strength comparison, no standard test.
- **Strength of aggregate** : should exceed that of concrete (rocks 70–300 MPa); weak aggregate (e.g. pumice) limits strength of concrete.

Test (IS 2386 Part 4)	Procedure	Limits (IS 383)
Aggregate Crushing Value (ACV)	12.5–10 mm aggregate in 150 mm cylinder, load 40 t in 10 min; % fines passing 2.36 mm	≤ 45 % (other concrete); ≤ 30 % wearing surfaces
Aggregate Impact Value (AIV)	10–12.5 mm, 14 kg hammer, 15 blows from 380 mm; % passing 2.36 mm	≤ 45 % ; ≤ 30 % for wearing surfaces
Los Angeles Abrasion	Aggregate + 6–12 steel balls, drum 500 rev (30–33 rpm); % loss passing 1.7 mm	≤ 50 % ; ≤ 30 % for wearing surfaces
Aggregate Abrasion (Deval), 10 % fines value	Rate of wear / load for 10 % fines	– (roads)
Toughness / Hardness	Impact & Abrasion tests	Toughness = resistance to impact; Hardness = resistance to wear

- $ACV = (W_2 / W_1) \times 100$, W_1 = weight of sample, W_2 = weight passing 2.36 mm.

11. PHYSICAL PROPERTIES OF AGGREGATES

11.1 Specific gravity, bulk density, porosity, absorption

- **Specific gravity** (G) = weight of dry aggregate / weight of equal volume of water. Normal rocks : **2.6 – 2.9** (granite 2.65, basalt 2.9). Apparent G (solid only) vs **bulk G (SSD)** (includes pores). Test : pycnometer / wire-basket method.
- **Bulk density** = weight / volume incl. voids: loose 1450–1550, compacted 1600–1750 kg/m^3.

Voids % = $(1 - \gamma / (G \cdot \gamma_w)) \times 100 \approx 35–40$ % for coarse.

- **Porosity** = volume of pores in aggregate / total volume : 0 – 20 %; affects strength, absorption, permeability, freeze-thaw.
- **Absorption** = $(W_{SSD} - W_{dry}) / W_{dry} \times 100$; 24 h immersion. Normal : 0.1 – 2 % ; > 2–3 % considered poor.

11.2 Moisture states of aggregate

State	Description
Oven-dry	All moisture removed (110 °C); absorbs water from mix
Air-dry	Surface dry, some internal moisture
SSD (saturated surface-dry)	Pores full, surface dry – **reference state** (no water exchange)
Damp / wet	Pores full + surface (free) moisture → adds to mixing water

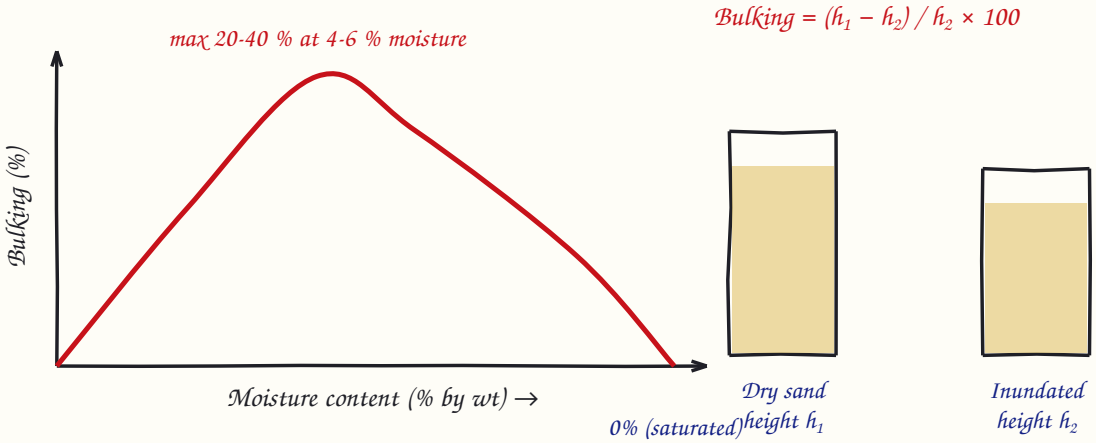
Moisture content & surface water

Moisture content (%) = $(W - W_{dry}) / W_{dry} \times 100$

Free (surface) moisture = Moisture content – Absorption (negative → aggregate absorbs water from the mix)

Correction : water to add = designed water – Σ(free moisture) + Σ(absorbed)

12. BULKING OF SAND



- Thin film of water on sand particles → **surface tension pushes grains apart** → increase in **volume** of moist sand. Fine sand bulks more (up to 40 %) ; coarse sand ≈ 15–20 %.
- Maximum bulking at **4–6 % moisture** ; when completely saturated (submerged) → bulking nil.
- **Field test** : fill a measuring cylinder (250 ml) with moist sand to height h_1 ; add excess water, stir – sand settles to h_2 → bulking % = $(h_1 - h_2)/h_2 \times 100$.
- **Importance** : if volume-batching is used without correcting for bulking, the sand actually

delivered is **less** → harsh, honeycombed, lower-strength mix. Hence quantity of moist sand is increased by the bulking %.

13. DELETERIOUS SUBSTANCES IN AGGREGATE

Impurity	Harmful effect	Limit / test
Organic matter (humus)	Retards hydration, loses strength	Colour test with NaOH — colour must not be darker than standard (straw/yellow)
Clay, silt, fine dust	Increases water demand, shrinkage, coats grains (poor bond)	Max 3 % (natural sand) , 1 % coarse (IS 383); sedimentation / decantation test
Salts (chlorides, sulphates)	Efflorescence, reinforcement corrosion, sulphate attack	Chloride ≤ 0.06 % ; total sulphate ≤ 0.5 % (as SO_3)
Coal, lignite, shale	Swelling, staining, popouts	≤ 1 %
Unsound particles (soft stone, mica, pyrites)	Weakness, rust stains, pop-outs	Max 5 % by wt
Sea-shells, sea salt (sea sand)	Chloride → corrosion	Must be washed

14. SOUNDNESS OF AGGREGATE

- **Soundness** = resistance of aggregate to **disintegration/volume change** caused by weathering – freezing–thawing, wetting–drying, heating–cooling.
- **Test (IS 2386 Part 5)** : sample immersed in saturated Na_2SO_4 (or $MgSO_4$) solution for 16–18 h → oven dried at 105–110 °C → **5 cycles**; salt crystallisation in pores simulates freeze–thaw.
- Limits (IS 383) : weight loss after 5 cycles **≤ 12 %** with sodium sulphate, **≤ 18 %** with magnesium sulphate.

15. ALKALI–AGGREGATE REACTION (AAR)

- Reaction between **alkalis** (Na_2O, K_2O) of cement (and other sources) and certain **reactive minerals** of aggregate in presence of moisture, producing **expansive gel** → cracks, pop-outs, map cracking, loss of strength & durability.
- **Alkali–Silica Reaction (ASR)** – reactive silica (opal, chalcedony, chert, tridymite, cristobalite, volcanic glass) + alkali → **alkali-silicate gel** which absorbs water and swells (pressure > tensile strength).
- **Alkali–Carbonate Reaction** – dolomitic limestone + alkali → de-dolomitisation, brucite.
- **Conditions** : (1) reactive aggregate (2) alkali > 0.6 % (Na_2O eq = Na_2O + 0.658 K_2O) (3) enough

moisture (RH > 80 %).

Control of AAR ★

- Use **non-reactive aggregates** (petrographic test, mortar-bar test ASTM C227, chemical test ASTM C289)
- **Low-alkali cement** (< 0.6 % Na_2O eq) ; use PPC / fly ash / slag / silica fume (they consume alkalis)
- Keep concrete **dry** (prevent moisture ingress), lithium admixtures, limit cement content.

16. THERMAL PROPERTIES OF AGGREGATES

- **Coefficient of thermal expansion** : 5 – 13 × 10^{-6} /°C (quartzite, flint highest ≈ 11–13 ; limestone, basalt ≈ 5–7 ; granite 7–9). Concrete ≈ 10 × 10^{-6} /°C (steel 12 × 10^{-6}) – compatibility of concrete & steel is the reason RCC works.
- Difference between aggregate and paste expansion → internal stress in fire / freeze-thaw cycles. **Specific heat** ≈ 0.18–0.22 cal/g°C ; **thermal conductivity** 1.5 – 3.5 W/m·K ; **diffusivity** governs cooling of mass concrete.
- Low-expansion aggregate (limestone) is preferred for fire resistance & pavements, high-expansion ones cause cracking.

17. SIEVE ANALYSIS, FM & GRADING

- **Grading** = particle size distribution of aggregate, found by **sieve analysis** (IS 2386 Part 1): a weighed sample is passed through sieves of decreasing size arranged in a nest and shaken ; % retained on each sieve is found.
- IS sieves : coarse – **80, 40, 20, 10, 4.75 mm** ; fine – **2.36 mm, 1.18 mm, 600 µ, 300 µ, 150 µ.**
- **Fineness Modulus (FM)** = Σ (cumulative % retained on standard sieves 80 mm … 150 µ) / 100. Higher FM = coarser.

Sieve size	Weight retained (g)	% retained	Cumulative % retained
10 mm	0	0	0
4.75 mm	40	4	4
2.36 mm	100	10	14
1.18 mm	150	15	29
600 µ	250	25	54
300 µ	250	25	79
150 µ	150	15	94

Sieve size	Weight retained (g)	% retained	Cumulative % retained
Pan	60	6	–

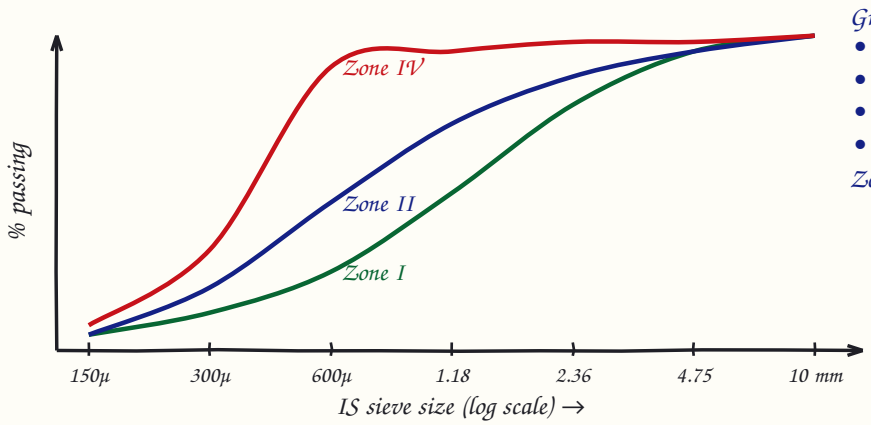
Worked example (1 kg sand) ★

FM = (0 + 4 + 14 + 29 + 54 + 79 + 94) / 100 = **2.74** → medium sand (Zone II).

Fine sand FM 2.2 – 2.6 ; medium 2.6 – 2.9 ; coarse 2.9 – 3.2 ; **coarse aggregate 5.5 – 8.0** ; all-in aggregate 3.5 – 6.5.

Combined FM of blend : $FM = (x{\cdot}FM_1 + y{\cdot}FM_2) / (x + y)$ [or % fine aggregate = $(F_{CA} - F_{blend})/(F_{CA} - F_{FA}) \times 100$]

17.1 Grading curves



Grading curve:
- S-shaped = well graded
- Straight/steep = uniform
- Hump = gap graded
- Zone I coarse → Zone IV fine

▶ **Well graded** (S curve) : all sizes present → min. voids, less paste, best strength & workability.
Gap graded : some intermediate sizes missing (suited for no-fines / pumping with care).
Uniformly graded : single size.

17.2 Grading of fine aggregate (IS 383 : 2016) – % passing

IS sieve	Zone I	Zone II	Zone III	Zone IV
10 mm	100	100	100	100
4.75 mm	90–100	90–100	90–100	95–100
2.36 mm	60–95	75–100	85–100	95–100
1.18 mm	30–70	55–90	75–100	90–100
600 μ	15–34	35–59	60–79	80–100
300 μ	5–20	8–30	12–40	15–50
150 μ	0–10	0–10	0–10	0–15

▶ Zone I (coarsest) → Zone IV (finest); Zone II is the most commonly used; Zones I–III OK for RCC, Zone IV (very fine) needs richer mix/ reduced fine aggregate.

17.3 Grading of coarse aggregate (IS 383) – % passing

IS sieve	40 mm graded	20 mm graded	40 mm single	20 mm single
63 mm	100	–	100	–
40 mm	95–100	100	85–100	100
20 mm	30–70	95–100	0–25	85–100
10 mm	10–35	25–55	0–5	0–20
4.75 mm	0–5	0–10	–	0–5

18. MAXIMUM SIZE OF AGGREGATE (MSA)

- **MSA** = smallest sieve on which ≥ 10 % retained (nominal max size = size of sieve passing 90–100 %).
- Larger MSA → less water & cement (smaller surface area) → economical, less shrinkage; but **limited by** :
  - ≤ ¼ of **minimum thickness / dimension** of member
  - ≤ **5 mm less than minimum clear spacing** between reinforcement bars; ≤ ¾ of cover (IS 456 cl. 5.3.3)
  - 20 mm typical for RCC; 40 mm for mass / PCC; 10–12 mm for thin / heavily reinforced sections.
- Increasing MSA from 20 → 40 mm : strength ↑ only in lean mixes; in rich mixes strength falls (smaller bond area, more bleeding at ITZ).

19. QUALITY OF MIXING WATER (IS 456 : 2000, cl. 5.4)

- Water used for mixing and curing should be **clean, free from oils, acids, alkalis, sugars, salts, organic matter**. **Potable water** is generally satisfactory.
- pH ≥ 6. Acidity : max 2 ml of 0.02 N NaOH to neutralise 100 ml water (phenolphthalein). Alkalinity : max 25 ml of 0.02 N H_2SO_4 (methyl orange).

Impurity (permissible limit, mg/litre)	PCC	RCC
Organic	200	200
Inorganic	3000	3000
Sulphates (as SO_3)	400	400
Chlorides (as Cl)	2000	**500**
Suspended matter	2000	2000

- Comparison test : 28-day strength of cubes with test water ≥ **90 %** of strength with distilled water ; initial setting time not less than 30 min and difference from control ± **30 min**.

- **Sea water** : acceptable for PCC (not for RCC / prestressed) – causes efflorescence, corrosion, dampness, increases early strength but ↓ 28-day strength by ~ 10–15 %.

Quick revision – Unit I ★

- Cement : C_3S (early), C_2S (late), C_3A (flash set, gypsum), C_4AF (colour) ; w/c min ≈ 0.38 ; Le Chatelier ≤ 10 mm ; Initial ≥ 30 min, final ≤ 600 min.
- Admixtures : accelerator ($CaCl_2$), retarder (sugar), AEA (3–6 % air), plasticizer 5–15 %, SP 20–30 %.
- Aggregate : FM sand 2.2–3.2, CA 5.5–8 ; bulking max 4–6 % moisture ; soundness 5 cycles, 12 % / 18 % ; MSA ≤ ¼ min dimension.